🔧 OpenPipeline Processing

> **Series:** OPLOGS | **Notebook:** 3 of 8 | **Created:** December 2025 | **Last Updated:** 01/28/2026

Configuring Pipeline Stages for Log Transformation

This notebook covers OpenPipeline processing stages: parsing, enrichment, metric extraction, event generation, bucket routing, and filtering.

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Prerequisites

- ✅ Access to a Dynatrace environment with log data
- ✅ OpenPipeline configuration permissions

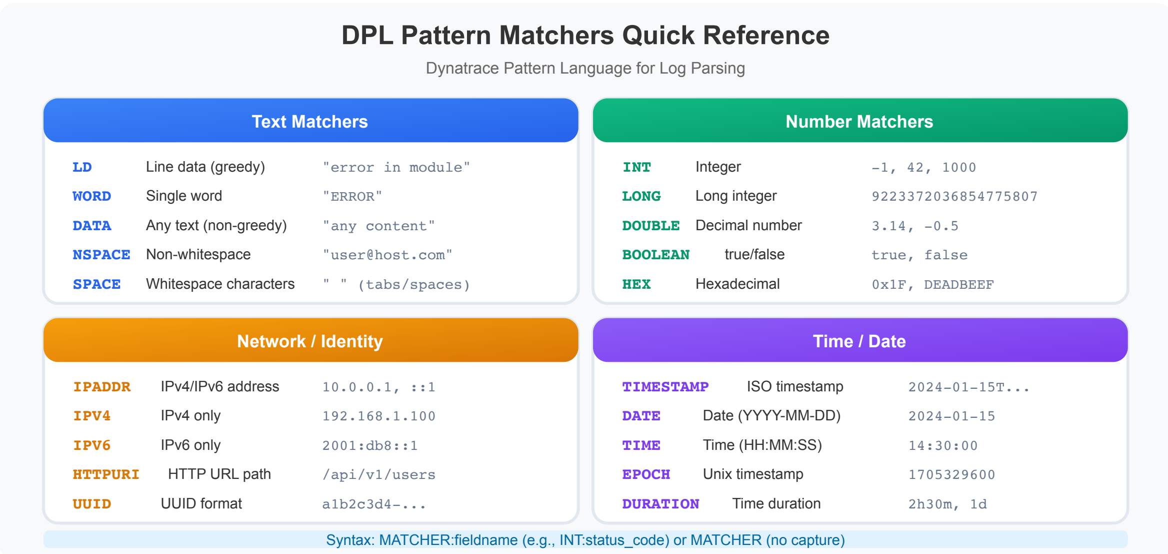
- ✅ Completed OPLOGS-01 and OPLOGS-02

> **OneAgent Attribute Enrichment (1.331+):** OneAgent can enrich all telemetry (metrics, spans, logs, events) with primary fields (`dt.security_context` , `dt.cost.costcenter`) and primary tags (`primary_tags.environment` , `primary_tags.team`) at the source. More efficient than auto-tags — feeds directly into OpenPipeline routing, bucket assignment, and Grail permissions. Configure via `oneagentctl --set-host-tag` or `--set-host-tag` at install time. See docs.

2. Parsing & Field Extraction

Parsing extracts structured fields from unstructured log content **at ingestion time**.

DPL (Dynatrace Pattern Language) Matchers



Matcher	Description	Example Match
`LD`	Line data (to delimiter)	Any text
`INT`	Integer	`42` , `-17`
`DOUBLE`	Decimal number	`3.14` , `-0.5`
`IPADDR`	IP address	`192.168.1.1`
`WORD`	Word characters	`hello123`

Matcher	Description	Example Match
`TIMESTAMP`	Date/time	Various formats
`JSON`	JSON object/array	`{"key": "value"}`

OpenPipeline Parse Processor

```
# Parse HTTP access logs
processors:
  - name: parse-http-logs
    type: dql
    source: content
    dql: |
      parse content, "IPADDR:client_ip SPACE LD SPACE '['
TIMESTAMP:request_time ']'
                      SPACE '\"' LD:method SPACE LD:path SPACE LD '\"'
                      SPACE INT:status SPACE INT:bytes"
```

```
// Discover log patterns for parsing design
fetch logs, from: now() - 1h
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 20
```

```
// Test parse pattern before configuring in OpenPipeline
fetch logs, from: now() - 1h
| filter contains(content, "GET") OR contains(content, "POST")
| parse content, "LD:method SPACE '/' LD:path SPACE INT:status_code"
| filter isNotNull(status_code)
| summarize {count = count()}, by: {method, status_code}
| sort count desc
| limit 15
```

```
// Verify fields already extracted by OpenPipeline
fetch logs, from: now() - 1h
| limit 100
| summarize {
    has_loglevel = countIf(isNotNull(loglevel)),
    has_status = countIf(isNotNull(status)),
    has_trace_id = countIf(isNotNull(trace_id)),
    has_span_id = countIf(isNotNull(span_id))
  }
```

3. Metric Extraction from Logs

Extract metrics with dimensions from log data at ingestion time. This is powerful for:
- Creating SLIs from log patterns
- Building dashboards without log queries
- Enabling metric-based alerting

OpenPipeline Metric Extraction

```
# Extract request duration metric from logs
processors:
  - name: extract-request-metric
    type: metric
    enabled: true
    condition: contains(content, "duration=")
    metricKey: log.request.duration
    dimensions:
      - service: k8s.namespace.name
      - method: extracted_method
      - status: extracted_status
    value: extracted_duration_ms
```

Common Metric Extraction Patterns

Log Pattern	Metric	Dimensions
Request completed	`log.request.count`	service, method, status
Response time	`log.response.duration`	endpoint, status_code
Error occurred	`log.error.count`	error_type, service

Log Pattern	Metric	Dimensions
Queue depth	`log.queue.size`	queue_name

```
// Identify logs with numeric values for metric extraction
fetch logs, from: now() - 1h
| filter contains(content, "duration")
        OR contains(content, "latency")
        OR contains(content, "time=")
        OR contains(content, "ms")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 15
```

```
// Simulate metric extraction: count by dimensions
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {
    request_count = count(),
    error_count = countIf(loglevel == "ERROR")
  }, by: {k8s.namespace.name, k8s.workload.name}
| fieldsAdd error_rate = round((error_count * 100.0) / request_count,
decimals: 2)
| sort error_count desc
| limit 15
```

```
// Preview: What dimensions would be valuable?
fetch logs, from: now() - 1h
| summarize {
    unique_namespaces = countDistinct(k8s.namespace.name),
    unique_workloads = countDistinct(k8s.workload.name),
    unique_pods = countDistinct(k8s.pod.name),
    unique_hosts = countDistinct(dt.entity.host),
    unique_sources = countDistinct(dt.openpipeline.source)
  }
```

4. Attribute Creation & Enrichment

Add **computed attributes** to logs for enhanced analysis and filtering.

OpenPipeline Field Processor

```
# Add computed attributes
processors:
  - name: enrich-environment
    type: fieldsAdd
    fields:
      - name: environment
        value: |
          if(contains(k8s.namespace.name, "prod"), "production",
          else: if(contains(k8s.namespace.name, "staging"), "staging",
          else: "development"))

      - name: severity_score
        value: |
          if(loglevel == "ERROR", 3,
          else: if(loglevel == "WARN", 2,
          else: 1))

      - name: team_owner
        value: |
          if(contains(k8s.namespace.name, "payment"), "platform-team",
          else: if(contains(k8s.namespace.name, "frontend"), "web-team",
          else: "unknown"))
```

Common Attribute Patterns

Attribute	Source	Use Case
`environment`	namespace name	Filter prod vs dev
`team_owner`	namespace/labels	Route alerts
`severity_score`	loglevel	Prioritization
`log_category`	content patterns	Classification

```
// Preview attribute creation logic
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| fieldsAdd environment = if(contains(k8s.namespace.name, "prod"),
"production",
                          else: if(contains(k8s.namespace.name, "staging"),
"staging",
                          else: "development"))
| fieldsAdd severity_score = if(loglevel == "ERROR", 3,
                             else: if(loglevel == "WARN", 2,
                             else: 1))
| summarize {count = count()}, by: {environment, severity_score, loglevel}
| sort environment asc, severity_score desc
```

```
// Categorize logs by content patterns
fetch logs, from: now() - 1h
| fieldsAdd log_category = if(contains(content, "Exception") OR
contains(content, "Error"), "exception",
                           else: if(contains(content, "request") OR
contains(content, "response"), "http",
                           else: if(contains(content, "database") OR
contains(content, "query"), "database",
                           else: if(contains(content, "auth") OR
contains(content, "login"), "security",
                           else: "general"))))
| summarize {count = count()}, by: {log_category, loglevel}
| sort count desc
```

```
// Identify namespaces for team ownership mapping
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {log_count = count()}, by: {k8s.namespace.name}
| sort log_count desc
| limit 20
```

5. Event Generation from Logs

Create **business events** from specific log patterns. Events flow to Grail and can trigger workflows.

OpenPipeline Event Processor

```
# Generate events from critical log patterns
processors:
  - name: generate-payment-events
    type: bizevents
    enabled: true
    condition: contains(content, "payment") AND contains(content,
"completed")
    eventType: com.example.payment.completed
    attributes:
      - payment_id: extracted_payment_id
      - amount: extracted_amount
      - currency: extracted_currency
      - customer_id: extracted_customer

  - name: generate-error-events
    type: bizevents
    enabled: true
    condition: loglevel == "ERROR" AND contains(content, "critical")
    eventType: com.example.critical.error
    attributes:
      - error_type: extracted_error_type
      - service: k8s.namespace.name
```

Event Use Cases

Log Pattern	Event Type	Purpose
Order completed	`order.completed`	Business analytics
User signup	`user.registered`	Funnel tracking
Deployment	`deployment.completed`	Change tracking
Critical error	`error.critical`	Incident trigger

```
// Discover patterns suitable for event generation
fetch logs, from: now() - 1h
| filter contains(content, "completed")
        OR contains(content, "success")
        OR contains(content, "failed")
        OR contains(content, "created")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 20
```

```
// Check existing bizevents (if any generated)
fetch bizevents, from: now() - 24h
| summarize {count = count()}, by: {event.type}
| sort count desc
| limit 15
```

```
// Preview: Logs that would become critical error events
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| filter contains(content, "critical")
        OR contains(content, "fatal")
        OR contains(content, "severe")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| fields timestamp, k8s.namespace.name, content_preview
| sort timestamp desc
| limit 20
```

6. Bucket Routing

Route logs to **appropriate buckets** based on content, source, or computed attributes.

Why Bucket Routing Matters

Bucket	Logs	Retention	Cost Impact
`default_logs`	Standard	35 days	Baseline
`debug_logs`	DEBUG/TRACE	7 days	80% savings
`audit_logs`	Security/compliance	365 days	Compliance

Bucket	Logs	Retention	Cost Impact
`error_logs`	Errors only	90 days	Investigation

OpenPipeline Route Processor

```
# Route logs to appropriate buckets
processors:
  - name: route-debug-logs
    type: route
    condition: loglevel == "DEBUG" OR loglevel == "TRACE"
    bucket: debug_logs

  - name: route-audit-logs
    type: route
    condition: contains(content, "audit") OR contains(content, "security")
    bucket: audit_logs

  - name: route-error-logs
    type: route
    condition: loglevel == "ERROR" OR loglevel == "FATAL"
    bucket: error_logs
```

```
// Current bucket distribution
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {dt.system.bucket}
| sort count desc
```

```
// Preview routing decisions
fetch logs, from: now() - 1h
| fieldsAdd target_bucket = if(loglevel == "DEBUG" OR loglevel == "TRACE",
"debug_logs",
                            else: if(loglevel == "ERROR" OR loglevel ==
"FATAL", "error_logs",
                            else: if(contains(content, "audit") OR
contains(content, "security"), "audit_logs",
                            else: "default_logs")))
| summarize {count = count()}, by: {target_bucket, loglevel}
| sort count desc
```

```
// Estimate storage savings from routing
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    total_logs = count(),
    total_mb = sum(content_bytes) / 1048576.0,
    debug_logs = countIf(loglevel == "DEBUG" OR loglevel == "TRACE"),
    debug_mb = sum(if(loglevel == "DEBUG" OR loglevel == "TRACE",
content_bytes, else: 0)) / 1048576.0
  }
| fieldsAdd debug_pct = round((debug_logs * 100.0) / total_logs, decimals: 1)
| fieldsAdd savings_7d_vs_35d = round(debug_mb * 0.8, decimals: 2)
```

7. Filtering & Sampling

Reduce log volume by **filtering** or **sampling** at ingestion.

OpenPipeline Filter Processor

```
# Drop noisy, low-value logs
processors:
  - name: drop-health-checks
    type: filter
    condition: contains(content, "health") AND loglevel == "INFO"
    action: drop

  - name: drop-heartbeats
    type: filter
    condition: contains(content, "heartbeat") OR contains(content,
"keepalive")
    action: drop

  - name: sample-debug-logs
    type: sample
    condition: loglevel == "DEBUG"
    rate: 0.1  # Keep only 10% of DEBUG logs
```

```
// Identify candidates for filtering (high volume, low value)
fetch logs, from: now() - 1h
| filter contains(content, "health")
        OR contains(content, "heartbeat")
        OR contains(content, "alive")
        OR contains(content, "ready")
| summarize {count = count()}, by: {k8s.namespace.name, loglevel}
| sort count desc
| limit 15
```

```
// Calculate potential savings from filtering health checks
fetch logs, from: now() - 24h
| summarize {
    total_logs = count(),
    health_logs = countIf(contains(content, "health") OR contains(content,
"heartbeat")),
    debug_logs = countIf(loglevel == "DEBUG")
  }
| fieldsAdd health_pct = round((health_logs * 100.0) / total_logs, decimals:
1)
| fieldsAdd debug_pct = round((debug_logs * 100.0) / total_logs, decimals: 1)
| fieldsAdd potential_reduction = round(((health_logs + debug_logs * 0.9) *
100.0) / total_logs, decimals: 1)
```

8. Complete Pipeline Example

Here's a comprehensive OpenPipeline configuration:

```yaml
# Complete OpenPipeline configuration for logs
name: production-log-pipeline
enabled: true

processors:
  # 1. FILTER - Remove unwanted logs first
  - name: drop-health-checks
    type: filter
    condition: contains(content, "health") AND loglevel == "INFO"
    action: drop

  # 2. PARSE - Extract structured fields
  - name: parse-http-logs
    type: dql
    condition: contains(content, "HTTP")
    dql: parse content, "LD:method SPACE '/' LD:path SPACE INT:status SPACE
INT:duration_ms"

  # 3. ENRICH - Add computed attributes
  - name: add-environment
    type: fieldsAdd
    fields:
      - name: environment
        value: if(contains(k8s.namespace.name, "prod"), "production", else:
"non-prod")

  # 4. MASK - Protect sensitive data
  - name: mask-emails
    type: mask
    field: content
    pattern: "[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\\.[A-Za-z]{2,}"
    replacement: "[EMAIL-MASKED]"

  # 5. EXTRACT METRICS - Create dimensional metrics
  - name: extract-request-count
    type: metric
    metricKey: log.http.requests
    dimensions:
      - namespace: k8s.namespace.name
      - status: status

  # 6. GENERATE EVENTS - Create business events
  - name: generate-error-events
    type: bizevents
    condition: loglevel == "ERROR"
    eventType: com.app.error
```

```
  # 7. ROUTE - Send to appropriate bucket
  - name: route-debug
    type: route
    condition: loglevel == "DEBUG"
    bucket: debug_logs
```

```
// Verify current pipeline processing
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    with_pipeline = countIf(isNotNull(dt.openpipeline.pipelines)),
    unique_pipelines = countDistinct(dt.openpipeline.pipelines)
  }, by: {dt.openpipeline.source}
| sort total_logs desc
```

```
// Pipeline processing summary
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {dt.openpipeline.pipelines,
dt.system.bucket}
| sort count desc
```

📝 Summary

In this notebook, you learned:

✅ **Parsing** - Extract structured fields at ingestion
✅ **Metrics** - Create dimensional metrics from logs
✅ **Attributes** - Add computed fields for analysis
✅ **Events** - Generate business events from patterns
✅ **Routing** - Direct logs to appropriate buckets
✅ **Filtering** - Drop/sample low-value logs

Key Takeaway

> **Process at ingestion, not at query time.** OpenPipeline processing is fundamental to cost optimization, data quality, and operational efficiency.

➡️ Next Steps

Continue to **OPLOGS-04: Buckets & Data Governance** to learn about storage management and retention policies.

📚 References

- OpenPipeline Overview
- OpenPipeline Processors
- Metric Extraction
- Event Generation
- DPL Pattern Language

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.